

TEAM MEXICO EDITION

DRIVER'S PLAYBOOK



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1. BACKGROUND

Being a Driver at FIRST is not just having control of the robot: it's having the responsibility to represent in the field the work of an entire team. Every move, every decision and every second counts. This manual is not designed to teach you how to move the joystick. It is designed to prepare you mentally, technically and strategically to compete at the highest level. Here begins the path to becoming a true driver of competition: precise, intelligent and ready to make decisions under pressure.

1.1 RESPONSIBILITY

Always keep in mind that being a Driver is perhaps the most important responsibility when it comes to the robot. A single minor mistake can cost the entire season's worth of work from the whole team. Remember that your teammates have placed their full trust in you, knowing you'll represent them well—and that kind of pressure is not something everyone can handle. Being your team's Driver is a unique and intense experience: when things go well, it's incredibly rewarding, but the weight of that responsibility is always there.

1.2 VALUES

As a member of the drive team you represent your team both on and off the field. If your team shirt is on, everyone expects nothing but the best from you. You are the people who interact with the other teams the most, working both with teams and against them. Always congratulate the teams you work with and are against respectfully. Here are some drive team core values. Review, understand, and use these while at competition.

1.2.1 HONOR

- Treat others as you would like to be treated, so it starts by respecting yourself too.
- Never underestimate the work of a team on the field, everyone has something to contribute and having an atmosphere of respect between the teams everything will flow better in the game.

- Always keep in mind that respect is not just deserved, it's earned.

1.2.2 UNITY

- Be open to different communication styles and adapt when working with diverse teams.
- Encourage quieter voices to speak up during strategy discussions, every perspective adds value.
- Recognize and celebrate the diversity of ideas, roles, and backgrounds that strengthen the FIRST community.

1.2.3 GRACIOUS PROFESSIONALISM

- Gracious Professionalism is at the heart of FIRST. It's a mindset that promotes excellence in everything you do while honoring others' contributions and treating everyone with dignity. In this spirit, intense competition and mutual respect go hand in hand. Compete with passion, but always act with kindness, fairness, and empathy. Success, learning, and collaboration should coexist in harmony.
- Always recognize that every victory is the result of your entire team's dedication. Celebrate your achievements together and extend that celebration to your alliance partners.
- Never forget that life is a carousel. One day you may be offering help, and the next you may need it. Be generous with support, advice, and encouragement, and treat all teams with the same respect you would hope to receive.

1.3 CONFIDENCE

- The best athletes in the world are the best because they are sure that they are the best, having confidence in yourself gives you calm in moments of pressure.

There is a slight line between being confident and the arrogance, be very careful to cross it, humility is the way to success

Know that there is always room for improvement, and that there is always someone out there faster, smarter, and stronger.

1.4 WELLNESS

If you have a healthy body therefore you will have a healthy mind, your whole body and soul is deeply connected, a quick run or 15 minute exercise may help you clear your head in the morning and give you drive for the rest of the day at competition. Having energy in competition is essential for better performance and this is achieved in the following ways.

1.4.1 SLEEP

- Competitions are high-pressure environments where staying calm and focused is essential. Lack of sleep can lead to irritability, poor concentration, and bad decision-making, which affects not only you but your whole team dynamic.
- From early mornings to late evenings, competition days are intense. Sleeping at least 6 hours ensures your energy levels stay high, reduces physical fatigue, and helps you stay engaged and motivated all day.

1.4.2 HYDRATE AND EAT PROPERLY

- Eating well gives you steady energy and keeps you alert throughout the day. Balanced meals that include complex carbohydrates, protein, fruits, and healthy fats help fuel your body without making you feel heavy or tired. Avoiding overly greasy or sugary foods is key to staying focused and feeling good.

- Staying hydrated keeps your body and mind functioning at their best. Dehydration can cause headaches, fatigue, dizziness, and slower reaction times—all things you want to avoid during an intense competition day. Drinking water regularly is essential, and electrolyte drinks like sports beverages (in moderation) can also help, especially in hot environments. Avoid too many sugary or caffeinated drinks, as they can dehydrate you more.

1.4.3 RESET

- During competition it is possible that you or one of the other members of the drive team will become frustrated. Remove yourself from the public eye and find a place to relax, take a moment alone to relax and reset, but make sure to return with a refreshed mindset, ready to compete at your best.

2. KNOWLEDGE

2.1 RULES

It's essential to know the game rules inside and out. Being familiar with every element and detail allows you to play confidently, without hesitation or fear of committing a penalty. This knowledge is crucial if you want to fully take advantage of the game, check for game manual updates and read the manual frequently to avoid forgetting some rules.

2.2 COMMUNICATION

Each year brings a new game, with new elements, field locations, and challenges. That's why it's essential to standardize the names for all key areas and components of the game. The entire drive team must agree on this shared nomenclature and use it consistently. Below are the elements that must be named together with your drive team to ensure clear and effective communication:

- Game piece
- Scoring location
- Feeding location
- Field position

2.3 GAMEPAD

Investing in a high-quality gamepad is key to consistent and reliable performance. As the main interface between the driver and the robot, the gamepad must respond quickly and accurately under pressure. A well-built controller reduces input lag, prevents unexpected disconnections, and feels more comfortable during long days.

It's also highly recommended to minimize the number of buttons assigned to the chassis driver. Ideally, their thumbs should stay on the joysticks at all times to ensure full control of movement. The systems driver can manage more buttons and complex functions, which allows for a better division of responsibilities. By keeping the chassis driver's layout simple, work with your software team to reduce the buttons and try to automate most of the robot actions, less buttons faster decisions.

Ask your software team to create a control map, especially if you're just starting out. Having a visual reference of what each button does helps you learn faster and teaches others how the controls are distributed. Below is an example of a gamepad map for reference. (Figure 1)

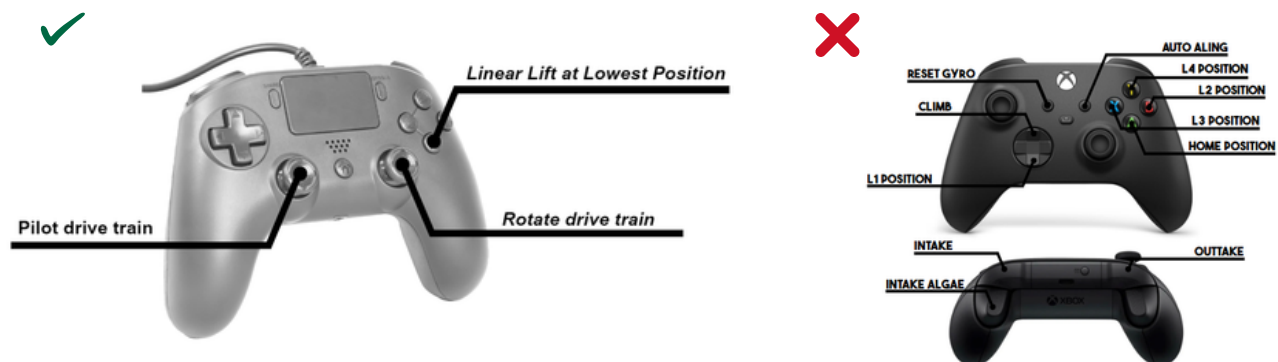


Figure 1. Gamepad map reference.

2.4 KNOW YOUR ROBOT

Understanding the performance limits of your robot is essential if you want to push it to its full potential. You need to know exactly what your robot can and cannot do, as well as its failure points. This knowledge not only helps you get the most out of your machine, but also allows you to quickly recognize when something goes wrong during a match. The better you understand how your robot behaves, the faster you can react when issues arise.

3. TRAINING

The best aren't the most naturally talented, they're the most dedicated. Becoming the best at anything requires hard work, consistency, and daily training. Remember, a mediocre robot can win with a great driver, but even the most advanced robot won't succeed with a mediocre one. The driver is the difference maker.

Get comfortable handling the robot and let go of the fear of breaking it during practice. This is the time to push the limits and discover what your robot can and can't do, and not discover it in the middle of a competition match. Remember, if you don't fail 100 times during practice, you risk failing 100 times in competition. Fail faster, learn faster.

Here are some skills you can practice to improve your control of the robot:

3.1 CHAIR DRILL (OMNI-DIRECTIONAL DRIVE)

Put chair on field, drive in circles around chair with one side of the robot tracking. The chassis should remain in continuous motion, no stopping and starting. (Figure 2)

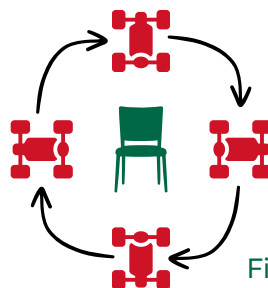


Figure 2. Chair drill example.

3.2 CONE SLALOM (TANK/OMNI-DIRECTIONAL DRIVE)

1. Make rows of cones.
2. Weave in between cones while spinning robot, if it is tank drive simply manage avoiding the obstacles and do it backwards. (Figure 2.1)

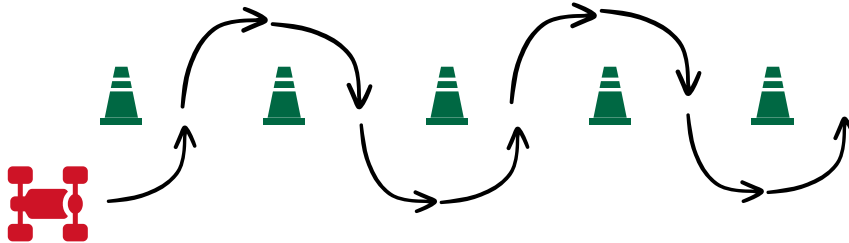


Figure 2.1. Cone Slalom example.

3.3 EVADE (TANK/OMNI-DIRECTIONAL DRIVE)

1. Put obstacles that make you dodge while you turn and move sideways or turn and move if it is tank drive.
2. Make it as smooth as possible, this drill is to avoid defense. (Figure 2.2)

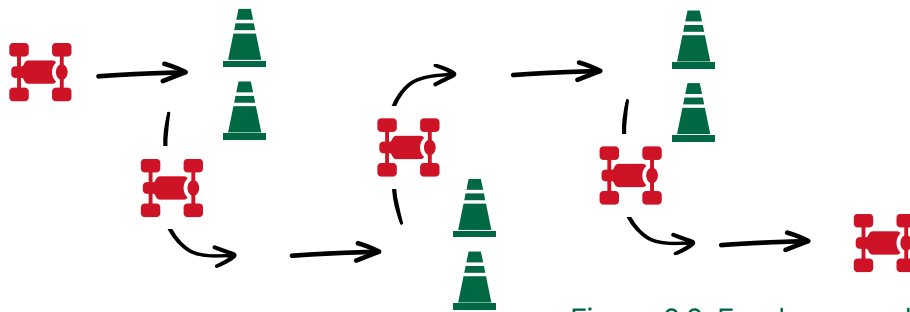


Figure 2.2. Evade example.

3.4 SPIRAL (TANK/OMNI-DIRECTIONAL DRIVE)

Drive in a spiral, gradually increasing the diameter of the circle with each rotation.

If you're using a tank drive, keep the robot facing forward during the spiral, and once you complete it, repeat the drill driving backward. If you're using an omnidirectional chassis, keep the front of the robot pointed toward the center so it rotates while moving in a spiral.

(Figure 2.3)



Figure 2.3. Spiral Drive example.

3.5 KEY ACTIONS

- In every challenge, try to use visual reference points on the field to help execute game tasks more accurately. For example, position the robot against a wall to shoot, or align it with a specific field element to pick up a game piece. These consistent points help improve precision and speed during matches.
- Remember smooth is fast. Start by driving at a slow, controlled pace and gradually increase your speed. The goal is to avoid choppy movement like sudden starts and stops that break your flow. It's better to move slowly but constantly.
- Identify the specific drills you need to practice for each challenge. Focus on the ones that allow you to score the most points in the least amount of time or help you organize your scoring more efficiently. For example, you can start by scoring in closer areas and gradually move to further one's as the match progresses. Defining these drills will help you maximize your scoring, remember, scoring is the most important objective in any challenge.

3.6 MATCH SIMULATIONS

Running practice matches is incredibly valuable. Simulating real match conditions including endgame or even the autonomous period, helps you get used to performing under pressure and allows you to continuously improve your cycle times.

Another helpful habit is to record and track your performance in each match. Writing down how many cycles you completed allows you to measure your progress over time. Watching the recordings helps you spot specific areas for improvement and make targeted adjustments to get better each time.

Sometimes you won't be able to practice with the real robot in the workshop—and that's where virtual simulators come in. Tools like XRC Simulator and MoSimulator replicate real-life robot physics as closely as possible, making them great options to keep training anytime, anywhere.

4. FIELD

4.1 SCORING TIPS

- Drive it like you stole it. Don't be afraid to push your robot to its limits—it's built to handle it. Many drivers hesitate because they feel responsible for keeping the robot in perfect condition. However, during a match, hesitation can cost valuable seconds or scoring opportunities. Trust your driving, rely on your training, and remember that the pit crew is there to repair any damage. Your focus should be on maximizing performance and driving assertively
- Don't waste time, every second matters. Matches are short and intense. Successful drivers plan ahead and move with purpose. Avoid indecision and minimize idle time. Smooth, continuous cycles can outscore faster but disorganized movements. Whether you're moving a game piece or transitioning between tasks, make every second count. Well-timed plays in the final moments of a match often determine the winner.

- Assign roles and divide the field. Before every match, define specific roles for each robot in your alliance. Whether it's a scoring bot, a defender, or a hybrid role, clarity prevents overlap and confusion. Dividing the field into zones helps each robot focus on their area and prevents interference. A well-coordinated alliance functions like a machine, each robot doing its part for maximum efficiency and field coverage.

5. MATCH CYCLE

5.1 PRE-MATCH

While one part of the team is performing the pre-match robot checklist, making sure everything is working properly, the strategy team should be preparing the plan for the upcoming match. Start by defining your team's individual strategy: what your robot will do, where it will go, and what objectives it will prioritize.

Once that's clear, move on to the joint alliance strategy. Meet with your alliance partners to understand their capabilities and agree on how the field will be divided and responsibilities shared. A great tip is to use a field map to visually plan each robot's role, this helps avoid confusion, ensures better coordination, and makes the most of each alliance member's strengths, like this one. (Figure 3)

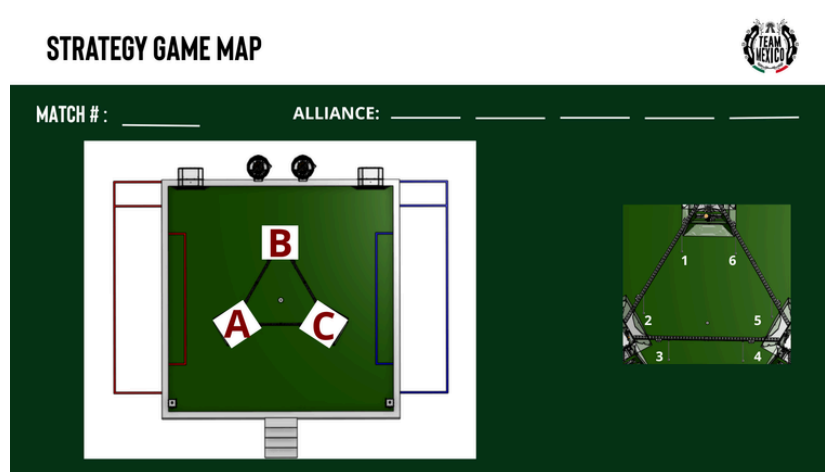


Figure 3

Then just before the match while you are in queue review the strategy with your whole alliance, be sure that everyone is on the same page, then now inside the field, it is highly recommended to do a Field checklist, a shorter checklist that will be the last chance to be sure if the robot is OK.

5.2 IN-MATCH

Once the match begins, there's no turning back. That's why it's so important to stay calm and focused, you'll make better decisions when your mind is clear. But above all, remember to enjoy the moment. When you're enjoying the experience, you're more likely to perform well. On the other hand, stress and fear of the outcome only increase the chances of mistakes. So breathe, trust your preparation, and have fun out there.

Remember to always push your robot to its limit, don't be afraid of breaking it. Taking your robot to its full potential is what leads to real performance and results. Playing it too safe holds you back. Trust the build, trust your training, and give it everything on the field.

Don't be afraid of making contact with other robots. It's completely normal to feel hesitant about collisions at first, but it's important to let go of that fear over time. Especially if you're playing defense, some level of physical contact is not only expected, it's part of the game. As long as it's within the rules, a little roughness is just another strategic element of competition.

5.3 POST-MATCH

After each match, it's essential to review the match video together with your drive team. Analyze any mistakes made and identify areas for improvement. Watching your performance helps you learn from experience, adjust your strategy, and build confidence. It also helps reduce stress, you'll clearly see what you did well and what you can improve for the next match.

6. ENJOY

Every match is a unique moment, one that you've earned through hours of preparation, effort, and teamwork. The sounds of the crowd, the pressure of the clock, it's something few get to experience. So live it fully. Enjoy every second behind the glass. Because at the end of the day, this is more than robots.

There's no better feeling than winning a match knowing you gave it everything. That satisfaction, that adrenaline, that pride, it only comes when you truly enjoy what you're doing. Let your passion drive you to improve, to lead, to be the difference. Be the driver your team believes in. And never forget, the ones who enjoy it the most are often the ones who win.

7. REFERENCES

This text is not the product of a single individual's idea; it was inspired by the following sources, which served as the foundation for its development.

1. Citrus Circuits. (2023, December 17th). Effective Driver Training in FRC - 2023 Fall Workshops [Video]. YouTube.
<https://www.youtube.com/watch?v=EeAovf8hZEs>
2. Team Rembrandts 4481. (2022). Drivers Manual 4481. ChiefDelphi.

